

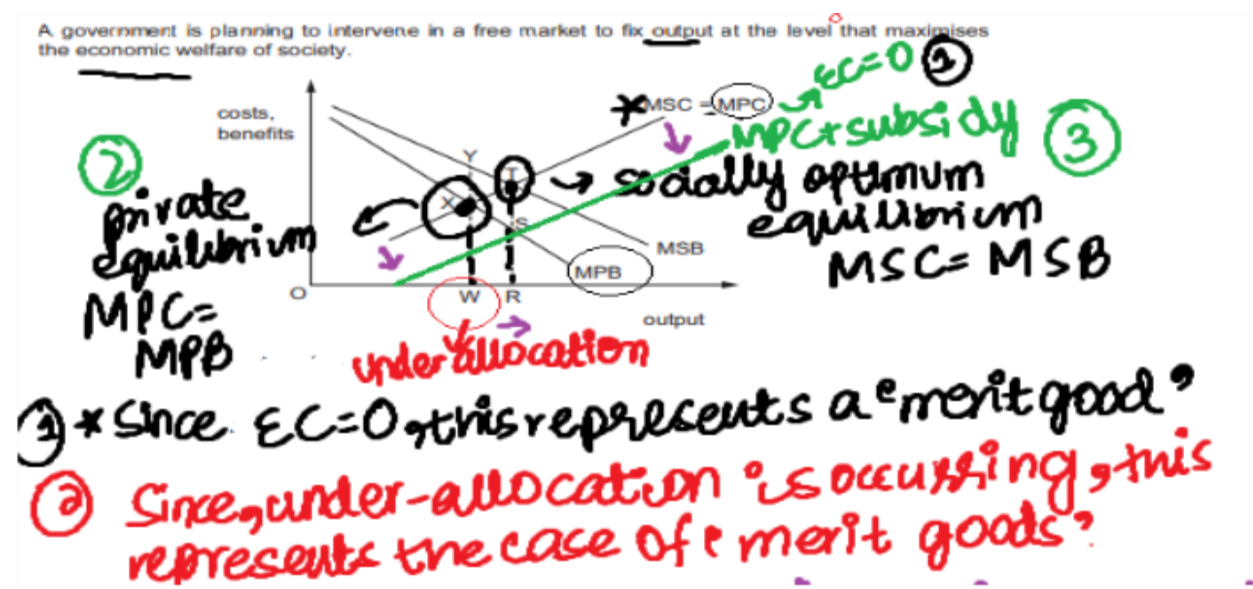
1-B:

In order for external costs or negative externalities to exist, the social costs must be greater than private costs.

- $SC = PC + EC$
- $SC - PC = EC$

This explains why option B will be correct.

2-A:



Why A and not B? Considering this under-allocation of merit goods, subsidy will have to be given in-order to increase output up to socially optimum level of output. The per-unit subsidy will be the vertical distance between the two supply curves (MPC) and (MPC+subsidy) i.e. "TS".

- Note: The vertical distance between the two supply curves is equal to "per-unit subsidy".
- And, the size of the external benefit quantifies the size of the subsidy.

This is why, key A, becomes correct.

Why not C or D? : Taxes will further reduce output, and it can be seen that already, there is "under-allocation" being done.

3-D:

Pareto Efficiency:

A situation is considered Pareto Efficient when no individual can be made better off without making someone else worse off. In other words, resources are allocated in such a way that it is impossible to improve one person's situation without deteriorating someone else's situation. This means that the allocation of resources is optimal, and no further improvements can be made without trade-offs

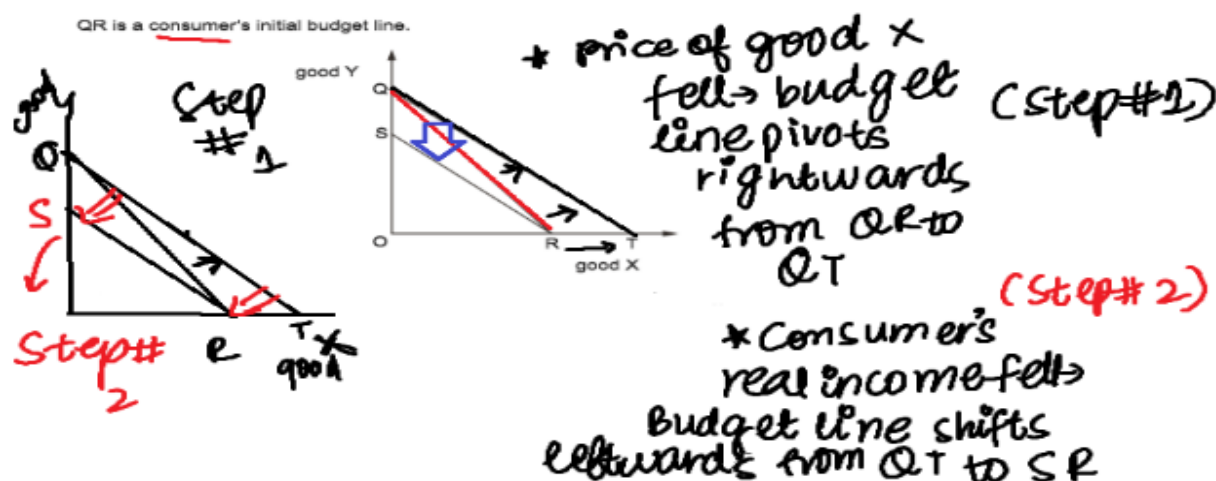
In this option we are interested in an option that results in “pareto improvement”

->A Pareto Improvement is a change that makes at least one individual better off without making anyone else worse off. In other words, it is a change that improves the situation of one or more individuals without deteriorating the situation of anyone else.

A,B,C: In all these three options , the income of X has been reduced to benefit Y with more income. This represents that pareto improvements have not taken place.

D: Similarly, in option D, the income of person X has been increased from 100 to 130, without person Y having to face a reduction in income. Since this represents pareto improvement, option D becomes correct.

4-A:



5-A:

C, D: Since all inputs can increase, this represents a long-run situation; law of diminishing returns and law of variable proportions represent short-run concepts. This is why, these options become incorrect

Why A and not B?

- ➔ When a firm increases all its inputs by 100% and its output increases by only 50%, this indicates that the firm is experiencing **decreasing returns to scale**. This means that the firm's output is not increasing proportionally to the increase in inputs, indicating a decrease in productivity.
- ➔ Economies of scale, on the other hand, occur when a firm expands its production and reduces its average cost per unit, often due to efficiencies and specialization. In this scenario, the **firm's output would increase by more than 100% when inputs are increased by 100%**, indicating increasing returns to scale.

6-C:

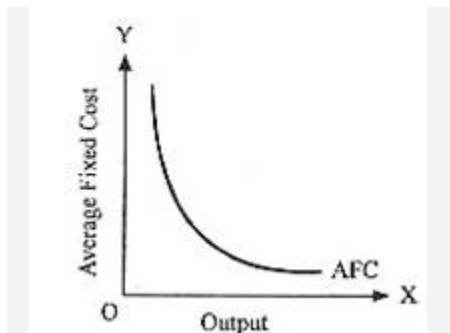
Quantity Sold	Price per Unit	TR=P×Q	AR= TR/Q	MR= TR _n -TR _{n-1}	
1	10	10	10	10	positive MR (elastic demand)
2	9	18	9	8	
3	8	24	8	6	
4	7	28	7	4	
5	6	30	6	2	unitary demand; TR unchanged
6	5	30	5	0	
7	4	28	4	-2	negative MR; inelastic demand

It can be seen above that when MR has a negative value, this represents inelastic demand, and when MR has a positive value, this represents elastic demand.

Similarly, the MR continuously decreases as price falls, and elasticity moves from elastic to inelastic, but this is in case of a linear demand curve, and not a unitary elastic demand curve.

But since, TR remains unchanged in case of unitary elastic demand, this represents that marginal revenue or additions to the total revenue is zero. This is why, option C is correct.

7-B:



Average Fixed Cost (AFC) falls as output increases because the fixed costs are spread over a larger number of units.

Let's use a numerical example:

Fixed Costs = \$100 (e.g., rent, salaries, etc.)

Initial Output = 10 units

$AFC = \text{Fixed Costs} / \text{Output} = \$100 / 10 = \$10 \text{ per unit}$

Now, let's increase output to 20 units:

$AFC = \text{Fixed Costs} / \text{Output} = \$100 / 20 = \$5 \text{ per unit}$

As you can see, the AFC has fallen from \$10 to \$5 per unit as output increased from 10 to 20 units. This happens because the fixed costs remain the same, but are now spread over a larger number of units, reducing the average fixed cost per unit.

8-A:

- ➔ A profit maximizing firm operates at output OW , because that is when, marginal cost is equal to marginal revenue. And since the last unit contributes as much as to the revenue as it does to the cost, this represents that profits will be maximized here. This is why, option A is correct

- ➔ At output OX, $MC=AC$, which represents a productively efficient level of output
- ➔ At output OY, $MC=AR$, which represents an allocatively efficient point
- ➔ At output OZ, $AC=AR$, which represents a normal profit equilibrium.

9-D:

- A- When $MR=MC$, this represents that a firm operates at a profit maximizing equilibrium
- B- When $AR=AC$, this represents a normal profit equilibrium, and not a revenue maximizing one.
- C- Marginal revenue is equal to average revenue in case of perfectly competitive firms when the price remains unchanged.

D-

B In-order to maximize Total Revenue, the marginal revenue must be "0", and this is shown in the diagram below. This is why, key **B**, will be correct.

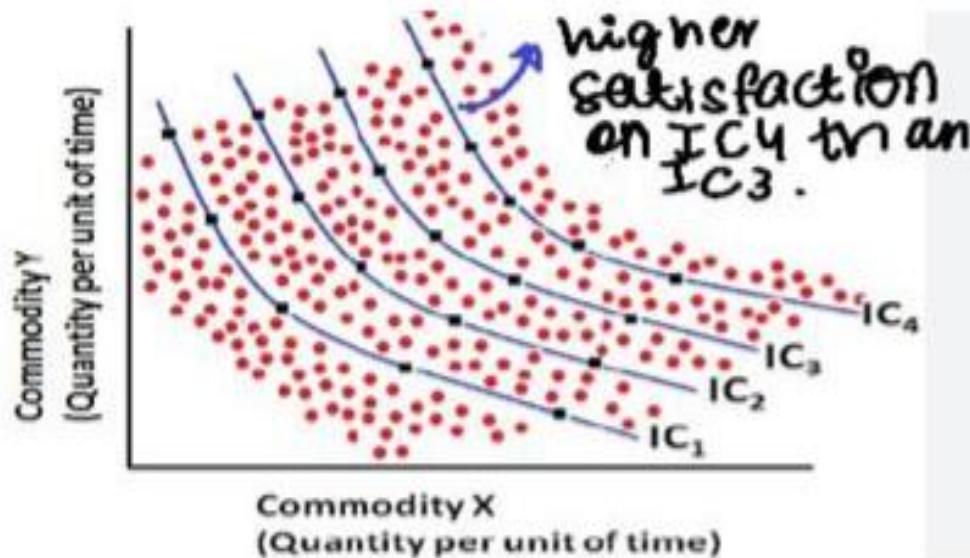


10-B:

A , C, D: These options are related to budget line. When prices change , the budget line tilts towards the right or left. And upon a change in incomes, the budget line shifts leftwards or rightwards parallelly.

B – This is correct because the further the indifference curve is from the origin, the greater will be the satisfaction that it will yield.

Therefore, a higher indifference curve "IC2" represents higher satisfaction than "IC1". This can be seen on the indifference map below:



11-B:

A – In a perfectly competitive market structure, there is no product differentiation being done. Instead, firms involve themselves in selling homogeneous products

B - Product differentiation is a major feature of monopolistic competition because firms try to differentiate their products from those of their competitors to gain a competitive advantage. This is done through various means such as: **1. Branding 2. Advertising 3. Packaging 4. Quality variations 5. Product features.** Because of this some firms face a higher demand than others in the short-run, thereby leading to them making more profits than others.

But in the long-run, competitors are attracted because of super-normal profits, and so, the share of market demand held by each firm falls due to a higher degree of competition, and so, normal profits are earned in the long-run. **This is why, this becomes correct.**

C – Although in a non-collusive oligopoly, firms engage in non-price competition, which includes product differentiation, they can still earn **super-normal profits in the long-run.**

D – In a monopoly, there is one firm with the highest market power, which means that it does not have to engage in any form of product differentiation because there are no rivals producing close substitutes.

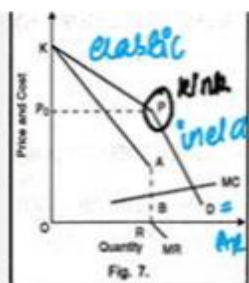
12-D:

A – The **kinked demand curve model** does not represent “collusion”, because collusion represents a situation whereby 2 or more firms in an oligopoly merge, secretly or openly, to form a “**cartel**”, to be able to **maximize “joint profits”**. In other words, because of this, an “**oligopoly turns into a monopoly**”.

B- **"Price leadership by a dominant oligopoly** is a market phenomenon where one firm, known as the dominant firm, takes the lead in setting prices, and other firms in the oligopoly follow its pricing decisions. This dominant firm has a significant market share and influence, often being the largest or most influential player in the industry. The other firms in the oligopoly are typically smaller and have less market power." **This is also not shown through the "kinked demand curve model"**

C- Oligopoly **does earn super-normal profits** in the long-run, but this is not represented through the “kinked demand curve model”

D - Because of the **reluctance of firms to alter prices**, prices remain **unchanged at the “kink”**. This is explained in detail below:



Price rigidity & Interdependence

The kinked demand curve is a theoretical model to understand how firms might respond to changes in prices

The key features of the kinked demand curve are as follows:

✿ **Flat and Elastic Range:** Above the kink, the demand curve is relatively elastic. It suggests that if the firm raises its price slightly in an hope to increase total revenues, it will lead to a significant decrease in the quantity demanded by consumers and subsequently, a fall in total revenue, because other firms will not follow, which represents "elastic demand" (since it is a behaviour of one firm) As a result, the firm would lose a substantial portion of its market share.

* Interdependence refers to the mutual influence and dependence among firms in an oligopoly market. Unlike perfectly competitive markets, where firms are price takers and have no impact on market prices, in an oligopoly, each firm's actions and decisions can significantly affect its competitors and the overall market. In the case of the kinked demand curve, firms in a non-collusive oligopoly closely observe and react to each other's pricing and output decisions. They are aware that any change in their price or production levels will trigger responses from their rivals. This interdependence means that firms must consider how their competitors might react to their actions.

Therefore, the prices remain "rigid" at the "kink".

This is because of the concept of "interdependence"

*** Steep and Inelastic Range:** Below the kink, the demand curve is relatively inelastic. It implies that if the firm lowers its price slightly, the change in the quantity demanded will be relatively small, or in other words, the qty demanded will extend by a relatively smaller percentage, because other firms in the industry will follow (representing "inelastic demand"), which results in a fall in total revenue. Competing firms are expected to follow the price reduction, resulting in losses faced by everyone.

As a result of which, key D, becomes correct.

13-D:

In both the countries, the concentration ratio between 2012-2015 was seen to rise.

$\uparrow$ US $\rightarrow$ 2012 $\rightarrow$ $43.0 + 3.3 + 50.6 = 96.9$
 $\rightarrow$ 2015 $\rightarrow$ $65.6 + 30.1 + 3.8 = 99.5$

13 The table shows percentage (%) market share of smartphone operating systems (OS) for Great Britain (GB) and the United States (US) in January 2012 and July 2015.

OS system	GB 2012 (%)	GB 2015 (%)	US 2012 (%)	US 2015 (%)
V	49.3	54.3	43.0	65.6
W	15.7	1.0	3.3	0.4
X	29.3	32.7	50.6	30.1
Y	2.4	12.0	2.1	3.8
Z	3.3	0.0	1.0	0.1

Handwritten calculations for CR3:
 2012: $49.3 + 15.7 + 29.3 = 94.3$
 2015: $65.6 + 30.1 + 3.8 = 99.5$

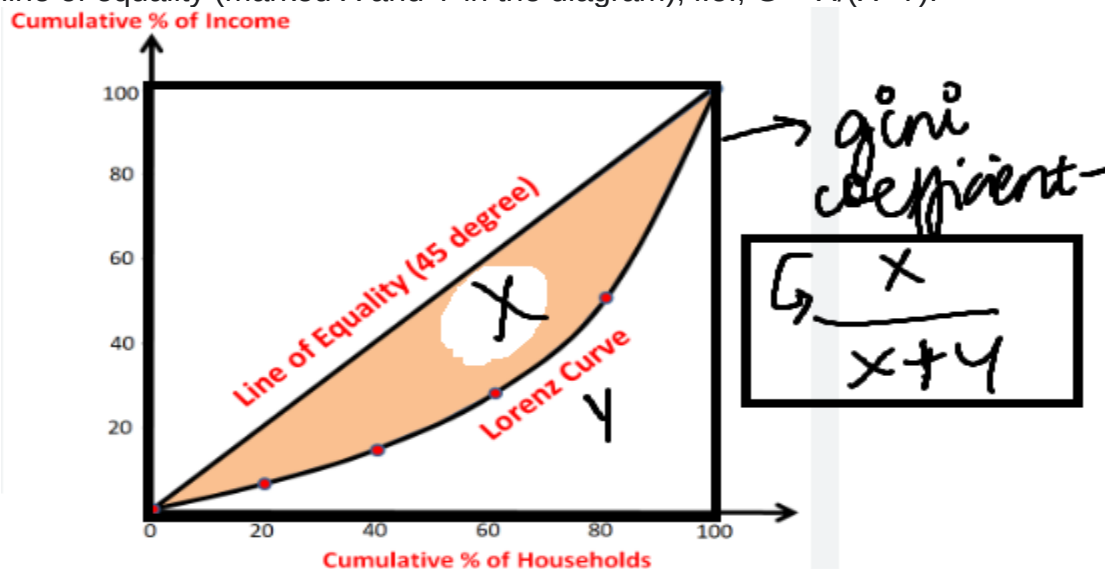
The three-firm concentration ratio is a measure of market concentration, which assesses the combined market share of the three largest firms in an industry. It is calculated by adding the market shares of the three largest firms and the results are expressed as a percentage.

The three-firm concentration ratio is defined as:

CR3 = (Firm 1's market share + Firm 2's market share + Firm 3's market share)

14-A:

The Gini coefficient can be thought of as the ratio of the area that lies between the line of equality and the Lorenz curve (marked X in the diagram) over the total area under the line of equality (marked X and Y in the diagram); i.e., $G = X/(X+Y)$.



15-C:

An increase in **tax-free income allowance** and an increase in **means-tested benefits** can lead to a more equitable distribution of income for several reasons:

1. Tax-free income allowance: **A tax-free allowance, also known as a personal exemption or basic allowance, is a portion of an individual's income that is not subject to income tax. In other words, it's an amount of money that you can earn without having to pay tax on it.**

- The government sets a threshold, say **\$12,000**, as the tax-free allowance.

Let's say Jane previously earned \$13000, this means that she was liable to pay tax on \$1000 earned above \$12000. Now when taxable allowance has been increased to \$15000, this represents that Jane will not have to pay any tax.

This is beneficial because it :-

- Reduces tax burden on low-income individuals, by allowing low-income earners to keep more of their income, thereby helping to lift them out of poverty

2. Means-tested benefits:

- Mean tested benefits target support to those who need it the most. This can be done by providing financial assistance to low-income individuals and families, through unemployment benefits for example. They therefore help to reduce income inequality by redistributing wealth.

16-D:

A – Marginal revenue productivity represents the additional revenue earned as a result of hiring an additional unit of labor.

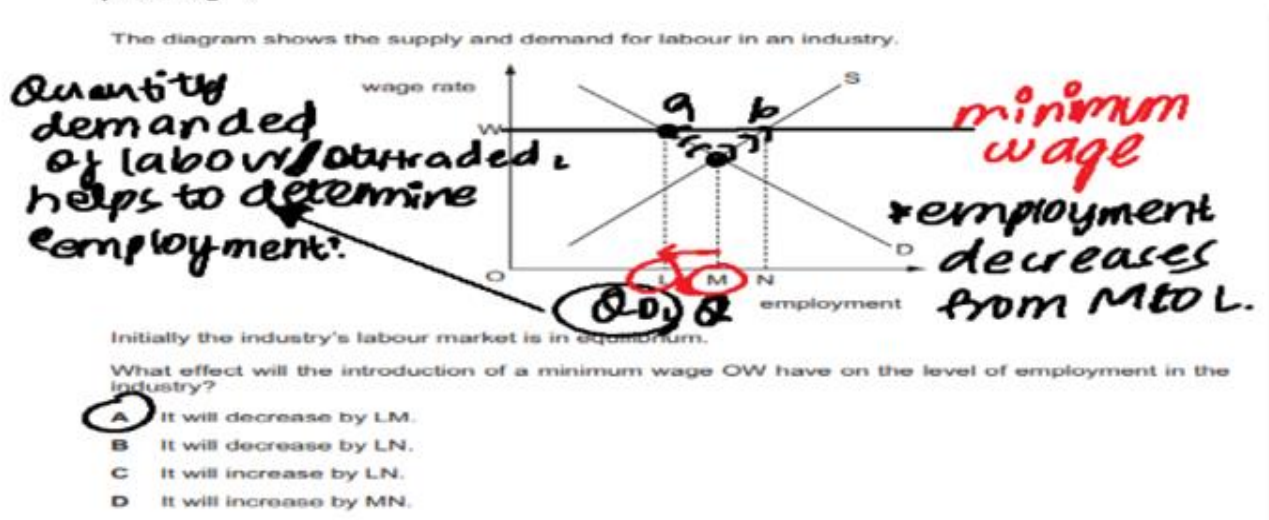
B – Producer surplus represents the difference between the minimum price a producer is willing to accept and the market price.

C and D - **Transfer earnings represent the minimum amount that needs to be paid to keep a factor of production in its present use. In this case, \$50 is the transfer earning because the shopkeeper would not be willing to work for any amount less than this. This explains why key D will be correct.**

Similarly, economic rent represents the amount that is paid in excess of transfer earnings. Since the shopkeeper is currently being paid \$75, when his transfer earning is \$50, this indicates that \$25 will be his “economic rent”

17-A:

→ Imposition of minimum price above the equilibrium wage rate would result in unemployment because **quantity demanded**, which represents the **equilibrium qty or qtv traded of labor**, thus helping to determine **“employment” contracts** *from M to L.*



18-C:

A – Based on the change in criteria, we cannot say that Canada intended to maintain the same level of skilled immigrants under both systems.

B – A fall in the weightage given to education under the new system represents that other factors may be deemed more important, for example: certainty of employment, but it does not represent that **Canada wanted a smaller number of highly educated skilled immigrants.**

C – Under the new system, half of the points (50 out of 100) are awarded because of certainty of job offer when previously this only accounted for 10 points. This represents that the certainty of employment has been made the main criteria, as a result of which, key C becomes correct.

D_A fall in the weightage given to work experience under the new system represents that other factors may be deemed more important, for example: certainty of employment, but it does not represent that **Canada wanted skilled immigrants to have a fewer year of work experience than previously.**

19-A:

A - The accelerator principle states that an increase in consumer spending (aggregate demand) leads to an increase in investment by firms. This is because when demand increases, firms need to expand production capacity to meet that demand, which requires new investment in capital goods such as equipment, buildings, and machinery.

This can be seen in the table below:

Case I. Life of the Machine 10 years.

	Period	Change in consumption	Capital Equipment needed	Gross Investment			Percentage change in Gross investment
				Additions	Replacement	Total	
10% rise in demand	0	1000	100	Nil	10	10	100% increase
	1	1100	110	10	10	20	

B A liquidity trap is a situation in which an increase in the money supply (e.g., through monetary policy) does not lead to an increase in consumption or investment. This occurs when interest rates are so low that they cannot fall further as at this point, money demand is perfectly elastic. This follows that people and businesses hoard cash, rather than investing or consuming. In a liquidity trap, aggregate demand is not responsive to changes in the money supply

C) The MEC represents the expected return on investment for a firm. It's the rate of return a firm expects to earn from investing in new capital goods (e.g., machines, buildings). When the MEC is high, firms are more likely to invest, as they expect a higher return on their investment.

D) The multiplier effect refers to the idea that an increase in spending (e.g., government spending or investment) leads to a multiplier effect on economic activity. The initial increase in spending creates additional rounds of spending, as the initial injection of funds is re-spent throughout the economy. The multiplier effect is often represented by the formula: $\Delta Y = k \times \Delta G$, where ΔY is the change in output, k is the multiplier, and ΔG is the initial increase in spending.

20-C:

multiplier: $\frac{1}{\cancel{mps} + \cancel{mpm} + \cancel{mpe} \rightarrow \text{no government}}$ $\rightarrow \frac{1}{1/6 + 1/3} \rightarrow 2$

21-A: Equilibrium is where:

$J=W$

$I+G+X = S+T+M$

21 The table refers to an open economy with a government sector.

In which situation is the economy in disequilibrium?

	C	I	S	G	T	X	M	J	W
A	400	150	80	150	200	120	100	420	380
B	450	150	100	150	230	120	90	420	420
C	480	150	110	150	210	120	100	420	420
D	520	150	120	150	210	120	90	420	420

22-C:

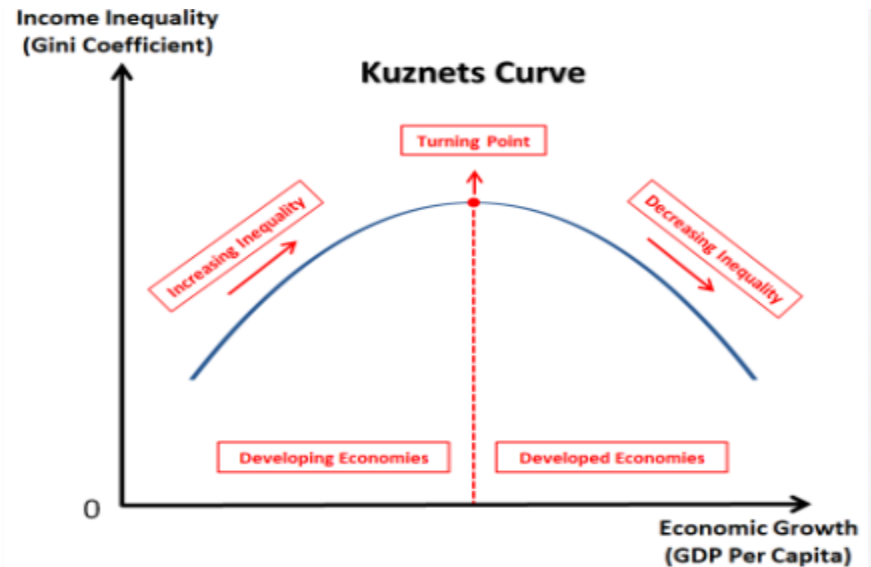
Commercial banks create money through a process called the money multiplier or credit creation. Here's how it works: A customer deposits money into their account at a commercial bank-> The bank is required to keep a fraction of this deposit as a reserve (e.g., 10%) -> The bank can lend out the remaining amount (e.g., 90%) to other customers->When the bank makes a loan, it credits the borrower's account, effectively creating new money->This new money can then be re-lent, creating even more money. This is why, option C is correct.

Cashing cheques for customers doesn't create new money; it just transfers existing money from one account to another. Increasing reserves at the central bank doesn't also create new money. Charging customers for banking services doesn't create money; it just generates revenue for the bank.

23-B:

On the vertical axis of the Kuznets curve, there is measure of inequality or in other words gini coefficient.

Whereas, on the horizontal axis, there is income per capita.



24-D:

Economic growth refers to an increase in a country's production capacity, typically measured by the increase in Gross Domestic Product (GDP) or Gross National Product (GNP). It focuses on quantitative expansion, such as an increase in GDP.

Economic development, on the other hand, encompasses not only economic growth but also social and structural transformations that improve the overall standard of living and well-being of a population. It involves qualitative changes that lead to improved education and healthcare, reduced poverty and income inequality, increased access to basic services like water, sanitation, and energy, enhanced human capital and institutional capacity, and promotion of social justice and human rights.

A – If economic growth results in rapid rural to urban migration, this will further lead to wages of the low-income groups being suppressed, thus further exacerbating inequalities.

B – If economic growth is based on extraction of mineral raw materials, it will lead to depletion, thus further giving existence to cost push inflationary pressures arising from shortages.

C – If economic growth mainly rewards entrepreneurs, it will not be inclusive and this widening of inequalities will worsen economic development.

D – If it results from the expansion of labor-intensive industries, this will create employment opportunities, which can lead to an increase in household incomes, improved standards of living, and reduced poverty. This represents that there will be an increase in economic development.

25-D:

A- It cannot be said that cost of living rose by 3.3% because in order to assess rises in cost of living, we need information on Consumer Price Index data.

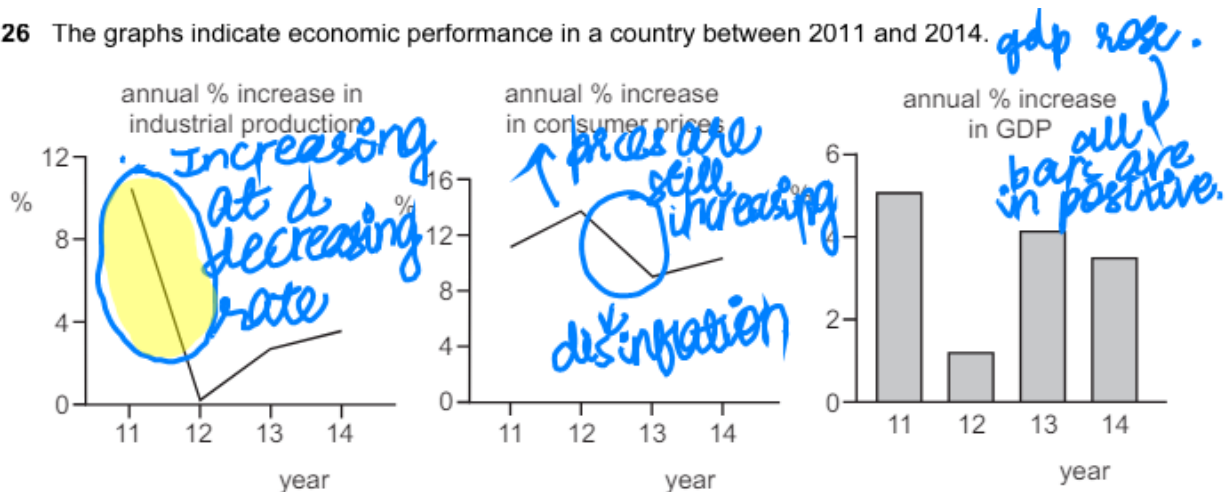
B- The overall output of goods and service rose, and not just of the industrial sector.

C- Whether or not inflation will occur because of rising Aggregate demand depends on the existence of spare capacity in the economy.

D- Since the economy expanded at a rate of 3.3%, this represents existence of positive economic growth.

26-D:

26 The graphs indicate economic performance in a country between 2011 and 2014.



In all the years, annual % increase in industrial production, annual % increase in consumer prices and annual % increase in gdp is in positive, which means that industrial production, consumer prices and GDP rose throughout.

Although in some years, the rate at which they rose fell, it still increased at a decreasing rate. This explains why option D will be correct.

27-B:

A – Rising unemployment will lead to a rise in the budget deficit of the economy. This is because, firstly tax revenues from incomes and expenditures will decline, and secondly, government expenditure on unemployment benefits paid to the unemployed increases.

B – A rise in unemployment represents a fall in purchasing powers of individuals, which will lead to a fall in consumption and subsequently Aggregate demand will decrease. Because of this, there will be a fall in inflationary pressures in the economy. As a result, key B, becomes correct.

C – A rise in unemployment represents a fall in purchasing powers of individuals, which will lead to a fall in consumption and subsequently Aggregate demand will decrease too. Because of this, **the actual output will fall.**

D – The economy's production possibility curve will not move inwards; instead, the actual will move closer to the origin.

28-C:

We are interested in an option which represents an **economy in a state of stability.**

A – When an economy was already facing high inflation, a rise in money supply through quantitative easing will lead to a rise in Aggregate demand, thereby leading to inflationary pressures being built up further.

B – In a situation, where there is already full-employment of the labor force, a further increase in Aggregate demand will lead to a positive output gap or inflationary gap.

C – When there existed a deep recession, an increase in Aggregate demand through the quantitative easing program will help to reduce the levels of demand-deficient unemployment, thereby leading to the economy being stabilized.

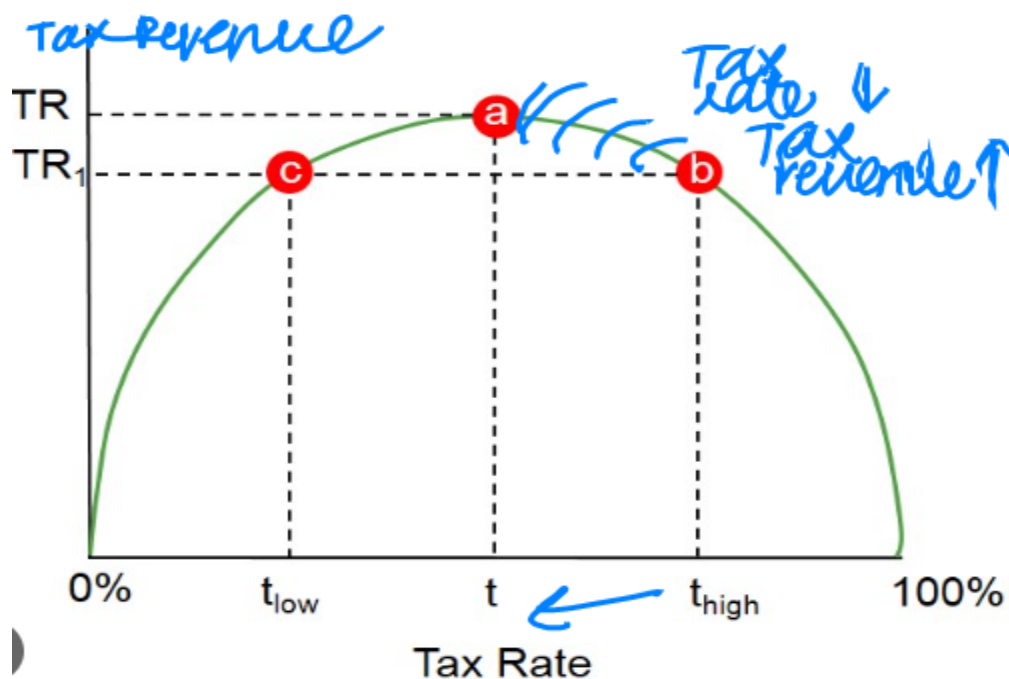
D – When because of a rise in real national incomes as a result of an increase in Aggregate demand, people have more money, this will lead to a further rise in import expenditures, which will worsen current account of balance payments deficit.

29-B:

This represents the Laffer Curve because it suggests that reducing the marginal tax rate can lead to an increase in tax revenue. The Laffer Curve is a theoretical representation of the relationship between tax rates and government revenue. It proposes that as tax rates increase, revenue initially increases, but eventually decreases as tax rates become too high and discourage economic activity.

In the UK example, the initial high marginal tax rate (83-98%) likely discouraged high earners from working, investing, or declaring income, resulting in lower tax revenue. By reducing the marginal rate to 40% , tax evasion falls, and the incentive to work, invest, and declare income increased, leading to an increase in tax revenue

This can be seen below:



30-C:

A – If there are high savings being done as a proportion of national income, this represents the fact that there are more funds available with the commercial banks to lend, and a rise in supply of loanable funds further brings about a fall in interest rates, which results in an increase in investment expenditure. As a result of this, the economic activity will be stimulated.

B – More efficient mechanisms for transmitting funds from savers to investors will encourage an increase in investment, thereby resulting in an increase in economic growth.

C – Detailed government planning will prevent rapid economic growth since planning in itself will be time-consuming, thus making this option correct.

D – More openness to foreign trade and investment will again stimulate the economy.

